

Priyanshu Tiwari

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Education

Jaypee Institute of Information Technology

Bachelor of Technology in Computer Science and Engineering

Sep 2022 – Sep 2026

Noida, India

Experience

India Space Lab

Embedded Systems Intern

Jun 2025 – Jul 2025

New Delhi, India (Remote)

- Built a **real-time drone tilt detection system** on an **ESP32-S3** with an **MPU6050** IMU, sampling **6 axes at 100 Hz** and driving OLED, LED and buzzer alerts the moment tilt crosses a **45 degree** safety threshold.
- Wrote **C++ firmware** that fuses accelerometer and gyroscope readings through a tuned **complementary filter** (alpha 0.96) into one stable attitude estimate, removing gyro drift and vibration noise using a timestep measured with **microS()**.
- Trained a **TensorFlow / Keras** neural network on labelled IMU data to classify safe versus dangerous tilt, then compressed it to a **2.5 KB TensorFlow Lite** model exported as a C byte array that fits in microcontroller flash.
- Designed and simulated the full circuit in **Wokwi**, with the IMU and a 128x64 OLED sharing one **I2C** bus plus three GPIO alert outputs, and documented wiring and thresholds so the build is **reproducible from the repository alone**.

Projects

PolyO | Python, PyTorch, tree-sitter, LightGBM, FastAPI, Docker, GitHub Actions

GitHub | Live

- Built an **ML service that predicts worst-case time and space complexity** of source code in **six languages** (Python, C++, Java, JavaScript, C, Go) just by reading it, with **no code execution** and **no LLM calls** at inference.
- Normalised all six languages into **one shared representation** with **tree-sitter**, so a single model serves every language. Trained on **194,000 labelled programs** with problem-level splits **enforced by a CI test** to block data leakage.
- Built and benchmarked **four models** on one held-out set, from a rule baseline up to a **graph neural network**. The GNN reached **0.377 / 0.336 macro-F1**, matching a zero-shot LLM on time and **beating it on space** at zero API cost.
- Shipped it as a **FastAPI** service in **Docker** with **GitHub Actions CI**, rewriting the GNN forward pass in **NumPy** to drop PyTorch from the image, verified identical to the trained model **within 1e-4**.

The Aerial Guardian | Python, PyTorch, YOLOv8, OpenCV, ONNX Runtime Web, JavaScript

GitHub | Live

- Built a **real-time multi-object tracking system** for drone video that runs **entirely client-side in the browser**, porting both detector and tracker to **ONNX Runtime Web** so the video never leaves the user's device.
- Stock YOLOv8 misses people in aerial footage because they are only **8 to 16 pixels tall**, so added a **high-resolution stride-4 detection head** to the network and retrained it on the **VisDrone** dataset on a single 6 GB laptop GPU.
- Implemented ByteTrack from scratch**: a **Kalman filter** for motion prediction plus **three-stage Hungarian matching** to hold object identities through occlusion, with **OpenCV** cancelling the drone's own camera motion.

Enagram.io | Next.js, TypeScript, PostgreSQL, pgvector, Gemini, OAuth 2.0

GitHub | Live

- Built a **conversational AI agent** that answers one question across a user's **Gmail, Drive and Calendar** in a single reply, linking the same person, project or file across all three services.
- Designed a **hybrid RAG retrieval layer** on **PostgreSQL + pgvector**, combining vector embeddings with keyword search so exact strings such as an invoice number are still found.
- Added **agent tool-calling** that creates and updates Calendar events and drafts Gmail replies inside the correct thread, all behind **Google OAuth** with per-user access scoping.

WeaponShield AI | Python, PyTorch, YOLOv8, ONNX Runtime, FastAPI, React

GitHub

- Built **real-time weapon detection** (pistol, rifle, knife) for CCTV, images and live webcam, fine-tuning **YOLOv8s** in two stages on **31,600 images** to reach **0.833 mAP50**.
- Served it through a **FastAPI** backend with **dual PyTorch / ONNX Runtime engines**, and cut false alarms with **K-of-N multi-frame confirmation** that alerts only when a detection persists across frames.

Technical Skills

Languages: Python, C++, JavaScript, SQL, Java

AI and ML: PyTorch, TensorFlow, YOLOv8, OpenCV, scikit-learn, LightGBM, Graph Neural Networks, Computer Vision, Object Detection and Tracking, RAG, Vector Search, LLM APIs (Gemini)

Backend and Data: REST APIs, Node.js, FastAPI, PostgreSQL, pgvector, MongoDB, NumPy, Pandas

Frontend: Next.js, React, Tailwind CSS, HTML5

Developer Tools: Docker, Git, GitHub Actions (CI/CD) basics, Linux, Postman

CS Core: Data Structures and Algorithms, DBMS, Operating Systems, OOP, Computer Networks

Achievements

- Competitive Programming: 600+ problems solved** on LeetCode (216 Easy / 317 Medium / **70 Hard**), contest rating **1533**, top **36% globally** across 11 rated contests. GitHub
- Hackathon Finalist:** advanced to the Final Round of **BEcon '24** at **IIT Delhi** among **30+ teams**.